
Learning the Tie-Line Interface: GNN Fast Screening and Learned Local P/Q Control for Generation Interconnection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Generation interconnection is an information problem: a planner must decide how
2 much power a new plant can inject through a single tie line without breaking voltage
3 stability or thermal limits on the bulk grid, and once connected, the plant’s smart
4 inverters must keep the grid within limits using only local measurements and one
5 broadcast signal. We build both ends of that loop on the IEEE 39-bus system with a
6 dedicated gen-tie line, measured under the full AC power-flow equations. A graph
7 neural network screens injection scenarios for congestion and stability headroom
8 at 6.2% RMSE on max line loading in distribution, with 91% congestion-decision
9 accuracy against 83% for a feature-based MLP, congestion-headroom ranking at
10 Spearman 0.75, and 0.55 ms per scenario against 256 ms per AC solve. Under
11 an N-1 topology shift, all three surrogate models we test fail, with loading errors
12 of 28–32% RMSE: a measured, reported negative. A learned local controller
13 maps local PCC voltage, output, and the screening signal to an active/reactive
14 dispatch inside the plant’s capability curve that, in the closed loop, reduces the
15 congestion-plus-voltage-plus-loss objective by 75% over the IEEE 1547 volt-var
16 baseline, with curtailment concentrated in exactly the scenarios the GNN flags as
17 congested. Failure modes are reported on both ends, and every number regenerates
18 from committed artifacts.

19 1 Introduction

20 The interconnection queue is the front door of the energy transition: in the United States, more than a
21 terawatt of generation capacity is waiting to connect, and the studies that decide whether a plant can
22 connect check thousands of candidate injections against the voltage and thermal limits of the bulk
23 transmission network [Gorman et al.(2025)]. FERC Order 2023 was written to make these studies
24 fast enough to be re-run as the queue changes [FERC(2023)]; ERCOT, which does not use FERC
25 procedures, has the same problem on the operating side, where plants behind weak tie lines get
26 curtailed because their real-time output congests the network [Gorman et al.(2025), ERCOT(2024)].
27 Both are one information problem: the boundary decision maker and the plant must know what the
28 plant’s power does to the whole grid.

29 Two gaps stand between the queue and the control room. Screening: full AC power flow is the
30 gold standard, but a single solve takes tens of milliseconds on a small system and seconds on a real
31 one, and a queue study runs thousands of them, while linearized DC screening is fast but blind to
32 voltage stability, the limit that binds on weak tie lines. Control: a connected plant with smart inverters
33 must regulate its active and reactive output (P and Q) against grid conditions, and the standard today
34 is a static volt-var curve from IEEE 1547-2018 [IEEE(2018)]: a fixed function of local voltage that
35 neither sees grid congestion nor adapts to the plant’s own headroom. Learned local control exists on
36 distribution feeders [Cui and Zhang(2022), Guo and Lei(2023), Liu et al.(2025)], but a controller is
37 only as good as the signal it conditions on: a purely local controller cannot know that a distant line is
38 congested.

This paper builds the two ends of the loop and measures them together. The first end is a graph neural network (GNN) that fast-screens injection scenarios for a gen-tie: given the network state and a candidate plant output, it predicts max line loading, minimum voltage, the loadability margin λ (voltage-stability headroom), and the congestion headroom κ (the plant-output multiplier before any line reaches 100% loading). GNNs are the established architecture because a power network is a graph and injection effects propagate along edges [Donon et al.(2020), Falk et al.(2019), Zhang et al.(2025)], evaluated for feasibility, constraint violations, and out-of-distribution (OOD) shifts per this workshop’s standard. The second end is a learned local controller: a small policy maps what the plant can measure locally (PCC voltage, its own real-time output) plus one broadcast signal from the screening module (predicted max loading) to a joint active/reactive dispatch, trained to imitate the grid-optimal dispatch found by a dense AC scan. Every policy is evaluated with the full AC power flow in the loop, since matching optimal setpoints in isolation does not guarantee loop stability.

Two measured results carry the paper. The first is a screening result with an honest negative: the GNN screens congestion decisions accurately in distribution (91% vs 83% for an MLP on the same features) and $470\times$ faster than one AC solve, but under an N-1 topology shift every surrogate fails (28–32% loading RMSE), which we report as a defined limit of surrogate screening rather than a solved problem. The second is a positive control result: the learned local controller reduces the congestion-plus-voltage-plus-loss objective by 75% over the IEEE 1547 baseline in the loop, keeping the improvement out of distribution. Failure modes are reported on both ends.

2 Problem formulation

2.1 AC power flow and the gen-tie model

A transmission network is a set of buses $i \in \mathcal{B}$ connected by lines and transformers. The AC power-flow equations balance net active and reactive injection at every bus,

$$P_i = \sum_j V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}), \quad Q_i = \sum_j V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}), \quad (1)$$

with V_i the voltage magnitude, $\theta_{ij} = \theta_i - \theta_j$ the angle difference, and G_{ij}, B_{ij} the conductance and susceptance of the line between i and j . A power-flow solution assigns voltages to every bus given the injections P_i, Q_i ; we solve (1) with the full Newton–Raphson solver of pandapower [Thurner et al.(2018)]. A scenario is *feasible* if the equations have a real solution in voltage limits; a non-converging AC solve, the practical signature of a voltage collapse, is infeasible.

The generation tie (gen-tie) is a dedicated line from a plant bus p to a point of common coupling (PCC) bus c on the bulk network. The plant is an inverter-based resource with active output $P_g \in [0, P_{\max}]$ and reactive output Q_g constrained by its apparent-power rating

$$Q_g^2 \leq S_{\max}^2 - P_g^2, \quad (2)$$

so the available reactive headroom $Q_{\max}(P_g) = \sqrt{S_{\max}^2 - P_g^2}$ shrinks as the plant produces more. Injecting P_g through the tie line changes every line flow and every voltage in the network; the screening question is how much it changes them and how much headroom remains.

2.2 Impact metrics

For a scenario with plant output P_g we compute four quantities under the full AC equations: *max line loading* $L_{\max}$ (largest line flow in percent of rating; congested if $L_{\max} > 100$); *minimum voltage* $V_{\min}$; the *loadability margin* λ , the largest factor by which loads and plant output can be scaled together before the AC power flow loses a solution, minus one; and *congestion headroom* κ , the largest factor by which plant output alone can be raised before any line reaches 100% loading, both computed by bisection with a full AC solve at every step.

2.3 Information structure for local control

The plant can measure its PCC voltage V_{pcc} and knows its own real-time output P_g , but not the state of the bulk grid. We give it one additional scalar, the screening signal ζ , the predicted max line loading from the GNN screening module, modeling an operator that broadcasts a single congestion indicator, as ERCOT-style curtailment signals already do [Gorman et al.(2025)]. The control problem is to choose the dispatch (P_g, Q_g) inside (2) as a function of $(V_{\text{pcc}}, P_g, \zeta)$ that keeps the grid inside its limits. The static baseline is the IEEE 1547 volt-var curve, reactive output as a fixed function of

model	max line loading RMSE (%)		
	in-dist	OOD load	N-1 topo
GNN	6.2 ± 0.3	10.6 ± 0.3	31.3 ± 1.8
MLP	11.0 ± 0.4	15.6 ± 1.5	28.0 ± 2.3
linear	4.9 ± 0.2	9.2 ± 0.0	31.8 ± 0.0

Table 1: Mean over four seeds of max line loading RMSE (%) for the GNN against an MLP and linear regression on the same node features, on the held-out training split, the OOD load regimes $\{1.45, 1.60\}$, and the N-1 topology. The GNN beats the MLP in distribution and under the load shift, but no model generalizes to the topology shift, a failure reported in the body. Congestion decision accuracy is reported in the body.

88 voltage with a ± 0.012 pu deadband and full capability at $|V_{pcc} - 1| = 0.06$ [IEEE(2018)]; a second
89 baseline holds a fixed 0.95 power factor.

90 3 Method

91 3.1 End 1: GNN fast screening

92 The network is a graph: buses are nodes, lines and transformers are edges. Node features are per-bus
93 active/reactive load and generation plus flags for the slack, plant, and generator buses; edge features
94 are resistance, reactance, susceptance, rating, tap ratio, and a line-versus-transformer indicator;
95 scenario-level plant output and load regime are appended as two extra node features. A three-layer
96 graph isomorphism network (GIN) [Xu et al.(2019)] with edge-feature message passing and sum-
97 pooling readout predicts $L_{\max}$, $V_{\min}$, λ and feasibility, trained with MSE plus binary cross-entropy;
98 κ is predicted by a second instance on the margin subset.

99 Training targets come exclusively from AC power-flow solves, so the model inherits the full physics
100 of (1) rather than a linearized approximation. Evaluation is constraint-aware: we report the accuracy
101 of the induced congestion decision ($L_{\max} > 100$), Spearman rank correlation of predicted versus
102 true loadability margin, and end-to-end inference time against the wall-clock time of one AC solve.

103 3.2 End 2: learned local P/Q control

104 The reference dispatch is computed per scenario by dense scan: we solve the full AC power flow over
105 a grid of Q_g values inside (2) and pick the reactive output minimizing the objective

$$J(P_g, Q_g) = \max(0, L_{\max} - 100)/100 + \max(0, |V_{pcc} - 1| - 0.012)/0.1 + 0.5 L_{\text{loss}}/P_{\text{load}}, \quad (3)$$

106 whose terms penalize congestion, voltage deviation, and losses (L_{loss} is network loss, P_{load} total load).
107 If the scenario is congested at the best reactive-only dispatch, we repeat the scan over curtailment
108 levels of active output and keep the jointly optimal pair (P^*, Q^*) .

109 A small multilayer perceptron policy π_ϕ maps the local observation (V_{pcc}, P_g, ζ) to a reactive
110 setpoint Q and a curtailment ratio $r \in [0.2, 1]$, trained by regression against the oracle targets
111 $(Q^*, P^*/P_g)$. The policy is evaluated the way it would deploy: the plant measures V_{pcc} , the policy
112 returns (Q, r) , the power flow is re-solved, and the loop repeats to a fixed point, three iterations.
113 This is the system-level test: a controller that matched setpoints in isolation could still fail here. The
114 learned controller is compared against the IEEE 1547 volt-var curve and a fixed 0.95 power factor,
115 both run in the same closed loop, plus the oracle, on held-out scenarios and OOD load regimes.

116 4 Experiments

117 4.1 Setup

118 All experiments use the IEEE 39-bus (New England) test system [Athay et al.(1979)] from pan-
119 dapower [Thurner et al.(2018)], augmented with a plant bus connected to bus 4 by a 15 km, 230 kV
120 gen-tie line rated at $S_{\max} = 1000$ MVA with $P_{\max} = 900$ MW. Scenarios randomize per-bus loads
121 ($\pm 15\%$ around a regime multiplier), redispatch the nine conventional generators ($\pm 10\%$), and draw
122 the plant output uniformly in $[0, 900]$ MW, at training load regimes $\{0.85, 1.0, 1.15, 1.30\}$ and OOD
123 regimes $\{1.45, 1.60\}$. The screening dataset has 3,000 training scenarios plus 500 per OOD split
124 (load regime and an N-1 topology); margins are computed by bisection on a stratified subset of 250
125 scenarios. The control dataset has 750 scenarios (150 OOD). Every number regenerates from the
126 repository scripts (Appendix A).

4.2 Result 1: the GNN screens injection scenarios accurately and cheaply

We train the GNN on 2,400 of the 3,000 training scenarios and evaluate on the held-out 600, the OOD load regimes, and the N-1 topology, over four seeds. The headline number is the accuracy of the induced congestion decision ($L_{\max} > 100\%$): 91% in distribution, against 83% for an MLP on the same node features (the linear model, given the regime multiplier as an explicit feature, reaches 93%). Loading itself is predicted with $6.2 \pm 0.3\%$ RMSE in distribution and $10.6 \pm 0.3\%$ under the OOD load shift (Table 1). End-to-end GNN screening of a scenario takes 0.55 ms against 256 ms for one AC solve, a factor of roughly 470, which is what makes re-screening the queue feasible.

The shift that matters for interconnection is structural: a line trips and the graph changes while the injections barely move. The N-1 split removes the most heavily loaded line (buses 15–18, 77.9% in the base case) and re-solves every scenario on the modified network. This is the paper’s honest negative: no surrogate generalizes (Table 1). The GNN, fed the modified graph, reaches 31.3% RMSE; the MLP and linear baselines, fed node features that are statistically identical between the two shifts, reach 28.0% and 31.8%. The topology change moves the loading distribution outside the training envelope and all three models underpredict it; the failure holds for the graph-structured model too, bounding what surrogate screening can promise without retraining.

Voltage-stability headroom is the harder target. On the 250 margin scenarios (bisection over full AC solves), the GNN’s congestion-headroom prediction correlates with the true injection multiplier at Spearman $\rho = 0.75$ ($n=200$ pooled held-out points), ranking which injections are closest to congesting the network. The loadability-margin prediction is weaker, $\rho = 0.31$: the model captures the broad trend of the margin surface (Figure 1) but not its sharp behavior near collapse, so the signal we broadcast to the controller is the congestion prediction.

4.3 Result 2: the learned local controller beats the static curve in the loop

Trained by regression against the dense-scan oracle on 600 scenarios, the policy is evaluated in the closed loop on 120 held-out plus 150 OOD scenarios, against the IEEE 1547 volt-var curve and the oracle, all in the same loop (Table 2, Figure 5). On held-out scenarios the learned policy reaches a mean objective of $J = 0.079$ against $J = 0.317$ for the volt-var droop, a reduction of 75%, kept under the OOD load regimes ($J = 0.605$ vs 0.846 for no control). The reduction comes from where the static curve is blind: the droop reacts only to local voltage, while the learned policy conditions on the broadcast screening signal ζ . It curtails real-time output in 43% of held-out scenarios, concentrated in the ones the screening module flags as congested (Figure 5b). The dense scan achieves $J = 0.031$, so the learned policy closes 83% of the gap between the static baseline and the grid-optimal dispatch.

4.4 Failure modes

We report failures, not only means (Figure 3). The headline failure is structural: all three screening models degrade to 28–32% loading RMSE under the N-1 topology shift (Result 1), so a surrogate screener must be re-validated or retrained when the topology changes, and flagged scenarios should defer to a full AC solve. The twelve worst held-out errors are high-loading 1.30-regime scenarios the model underpredicts. The learned controller is worse than the static droop on 15% of held-out scenarios, where the GNN underpredicts congestion and the policy curtails less than it should; flagged cases get a human or slower-solver review.

5 Discussion and limitations

The two results are two ends of one loop: the screening module broadcasts *one scalar*, predicted max line loading, and the plant’s policy turns that scalar plus its local voltage into an action that measurably reduces congestion, voltage deviation, and losses. What this paper does not claim is stated plainly. The 39-bus system with one gen-tie is a laboratory grid. The screening signal is reliable for congestion, not yet for stability headroom ($\rho = 0.31$); broadcasting it as a stability indicator would overclaim. Surrogate screening does not generalize to the N-1 topology shift we measure; generalizing without retraining remains open. The closed-loop controller is evaluated to a fixed point of the static equations, not against transient dynamics.

6 Broader impact

The concrete pathway is the interconnection queue: a screener that runs in milliseconds lets an operator re-run the study as the queue changes, and a controller using a broadcast congestion signal lets a connected plant manage output against real-time conditions. The honest stage matters as much: this is a 39-bus laboratory result, not authorization to deploy learned control on a live system, and the deployment envelope is exactly the failure modes we report (Section 4.4): flag-and-defer for the screening tail, with a fallback to the static curve when the signal is unreliable.

Use of generative AI

The manuscript was drafted with the assistance of a large language model; all experiments, code, and figures were produced by the authors’ own pipeline, per the AI4PowerGrids call for papers.

References

- [Athay et al.(1979)] Athay, T., Podmore, R., and Virmani, S. A practical method for the direct analysis of transient stability. *IEEE Trans. Power Appar. Syst.*, PAS-98:573–584, 1979.
- [Cui and Zhang(2022)] Cui, W. and Zhang, B. Lyapunov-regularized reinforcement learning for power system transient stability. *IEEE Control Systems Letters*, 6:974–979, 2022.
- [Donon et al.(2020)] Donon, B., Clément, R., Donnot, B., Marot, A., Guyon, I., and Schoenauer, M. Neural networks for power flow: graph neural solver. *Electric Power Systems Research*, 189:106547, 2020.
- [Falk et al.(2019)] Falk, T., Donon, B., Guyon, I., Donnot, B., and Schoenauer, M. Warm-starting AC optimal power flow with graph neural networks. In *NeurIPS 2019 Workshop on Tackling Climate Change with Machine Learning*, 2019.
- [FERC(2023)] Federal Energy Regulatory Commission. Improvements to generator interconnection procedures and agreements. *Order No. 2023*, Docket RM22-14-000, 2023.
- [ERCOT(2024)] Electric Reliability Council of Texas. 2023 state of the grid report. ERCOT, 2024.
- [Gorman et al.(2025)] Gorman, W., et al. Grid connection barriers to renewable energy deployment. *Joule*, 9(2):101803, 2025.
- [Guo and Lei(2023)] Guo, Y. and Lei, Q. Reinforcement learning-based smart inverter control with polar action space. *IEEE Trans. Power Systems*, 38(4), 2023.
- [IEEE(2018)] IEEE. IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces. *IEEE Std 1547-2018*, 2018.
- [Liu et al.(2025)] Liu, Y., Guo, Y., and Xu, T. Robust deep reinforcement learning for inverter-based volt-var control in partially observable distribution networks. *Applied Energy*, 2025.
- [Marot et al.(2021)] Marot, A., Donnot, B., Chaouache, K., Kelly, A., Dulac-Arnold, C., et al. Learning to run a power network challenge for training topology controllers. *Electric Power Systems Research*, 2021.
- [Okoyomon et al.(2025)] Okoyomon, E., et al. A framework for assessing the generalizability of GNN-based power flow solvers. In *e-Energy 2025*, ACM, 2025.
- [Thurner et al.(2018)] Thurner, L., et al. pandapower: an open-source python tool for convenient modeling, analysis, and optimization of electric power systems. *IEEE Trans. Power Systems*, 33(6):6510–6521, 2018.
- [Xu et al.(2019)] Xu, K., Hu, W., Leskovec, J., and Jegelka, S. How powerful are graph neural networks? In *International Conference on Learning Representations*, 2019.
- [Zhang et al.(2025)] Zhang, Y., et al. Operational risk quantification of power grids using graph neural networks. *Energy and AI*, 2025.
- [Zhou et al.(2024)] Zhou, X., et al. A power grid benchmark dataset for graph neural networks. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2024.

A Reproduction statement

Every number in this paper is regenerated from committed artifacts in the project repository: the scenario generator (experiments/make_screening_data.py, experiments/make_control_data.py), the trainers (experiments/train_screening.py, experiments/train_control.py), and the figure script (experiments/generate_figures.py). The data files and per-seed result JSONs are committed. All power-flow quantities come from pandapower AC solves; the loadability and congestion margins are bisection results over full AC solves; the control oracle is a dense scan over the reactive capability curve with full AC solves; the baselines are run in the closed loop to a fixed point. The plant is modeled as a constant-voltage resource in the screening study (a plant holding its terminal voltage) and as a PQ injection in the control study, where reactive dispatch is the control variable; a generator with a fixed voltage setpoint would ignore the reactive schedule entirely. Compute is a single CPU workstation; typical wall time for the full pipeline is a few hours.

B Additional figures and tables

B.1 Screening performance

B.2 Closed-loop control results

B.3 Per-seed screening results

All screening numbers are means over seeds; the per-seed values below are reported for completeness. Congestion decision accuracy is the fraction of scenarios where the model’s predicted max loading is on the correct side of the 100% threshold.

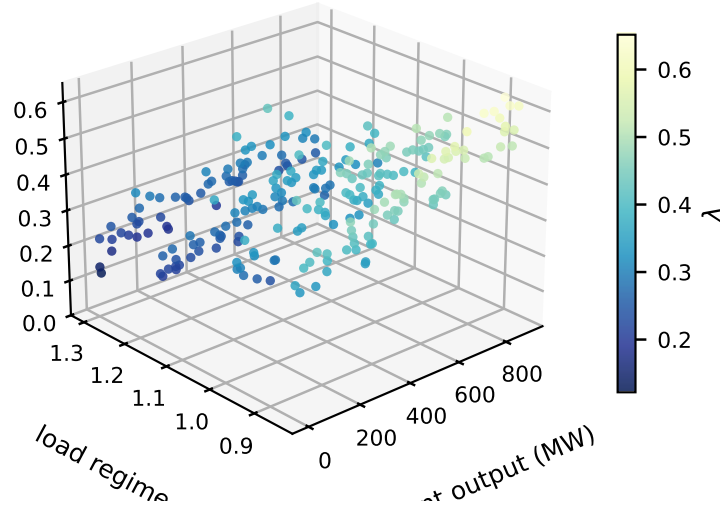


Figure 1: Loadability margin λ over the plant output and the load regime (250 scenarios, exact bisection over full AC power flow). The margin falls as the plant output rises and as the load regime rises; the surface is smooth in both directions, which is what lets the GNN regress it from a modest number of scenarios.

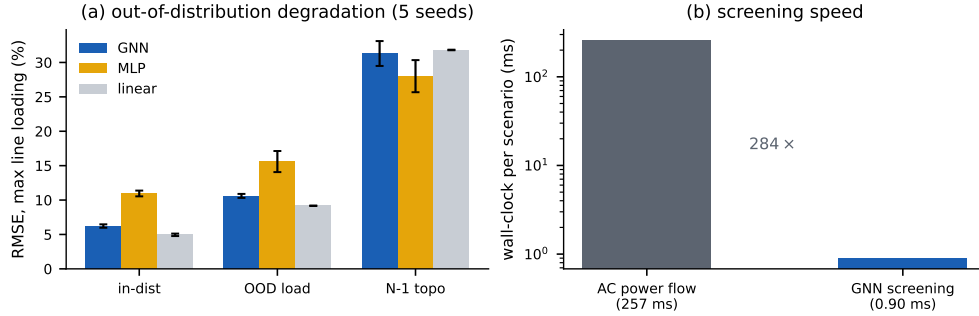


Figure 2: Left: per-split max line loading RMSE for the GNN, the MLP, and linear regression (mean and standard deviation over four seeds); the GNN beats the MLP in distribution and under the load shift, and all three models degrade to 28–32% under the N-1 topology shift. Right: wall-clock time per scenario for one full AC power-flow solve against end-to-end GNN screening inference (four-seed mean; log scale).

239 B.4 Per-seed control results

240 B.5 Training details

241 The screening GNN has three GIN layers of width 96 with edge-feature message passing (GINE),
 242 sum-pooling readout, and a three-output head (max loading, min voltage, feasibility logit); a second
 243 instance with a three-output head on the margin subset predicts loadability margin, congestion
 244 headroom, and feasibility. It is trained for 70 epochs (50 for the margin head) with Adam at learning
 245 rate 3×10^{-3} , cosine decay, weight decay 10^{-5} , and batch size 256, on MSE plus binary cross-entropy.
 246 The MLP baseline is two hidden layers of width 96 on the same flattened and normalized node features
 247 with the same optimizer. The linear baseline is ordinary least squares on the same features. The
 248 control policy is a two-hidden-layer MLP (width 64) mapping (V_{pcc}, P_g, ζ) to a normalized reactive
 249 setpoint ($\tanh$, rescaled by the capability curve) and a curtailment ratio (sigmoid, rescaled to $[0.2, 1]$),
 250 trained for 200 epochs with Adam at 10^{-3} , batch 128, MSE regression against the dense-scan oracle
 251 targets. The closed-loop controller and all baselines are iterated to a fixed point with three full AC
 252 solves. All models run on CPU; the scenario generators use eight multiprocessing workers.

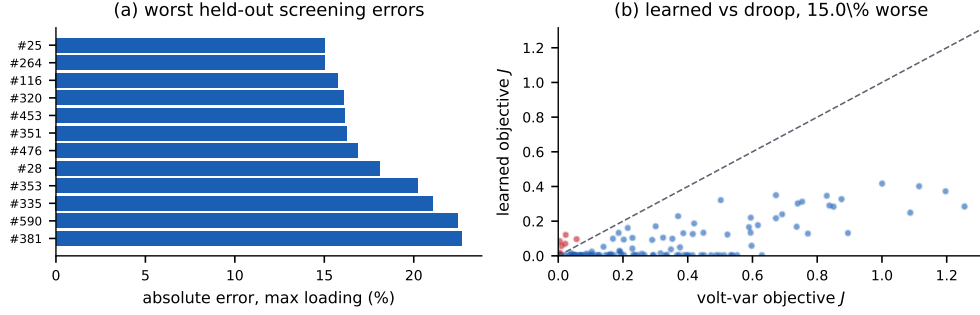


Figure 3: Left: the twelve worst held-out screening errors, the scenarios where the surrogate is least reliable and where a human review step should be triggered. Right: closed-loop objective of the learned controller against the IEEE 1547 volt-var baseline per held-out scenario, with the scenarios where the learned controller is worse shown in red.

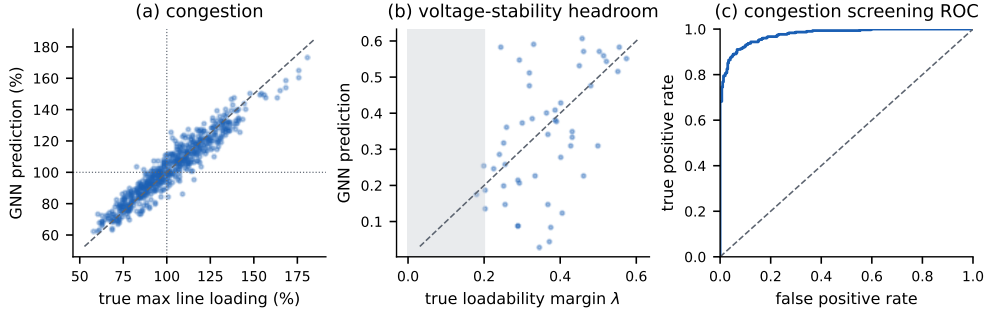


Figure 4: Screening performance of the GNN on held-out scenarios ($n=600$, 4 seeds) under the full AC power flow. Left: predicted versus true max line loading; the 45-degree line is exact prediction, points above 100% on both axes are congested scenarios. Middle: predicted versus true loadability margin, with the near-collapse region ($\lambda < 0.2$) shown; Spearman rank correlation reported in the body. Right: congestion screening ROC (any line above 100% loading) with the random baseline. Baselines: an MLP on the same node features and linear regression (Table 1).

Domain evaluation checklist

- Physics feasibility.** All training targets, the control oracle, and every closed-loop evaluation come from full Newton–Raphson AC power-flow solves (pandapower), never a linearized approximation. Constraint statistics are reported, not only aggregate error: congestion is defined by the induced decision accuracy on $L_{\max} > 100\%$ (body, Result 1 and Table 1) and voltage deviation by the exact margin violation rate in the control loop (Result 2).
- Out-of-distribution evaluation.** Both components are evaluated on two distribution shifts not seen in training: load regimes at multipliers $\{1.45, 1.60\}$ above the training range $\{0.85, 1.30\}$, and an N-1 topology with the most heavily loaded line removed. The screening models’ degradation on the N-1 shift is reported as a primary failure mode (Section 4.1, Results 1–2, Figure 2).
- Failure modes.** Reported alongside aggregate metrics: worst-case screening errors and the scenarios where the learned controller underperforms the static volt-var baseline (Section 4.4, Figure 3).
- System-level effect.** The learned controller is not scored on setpoint error. It is placed inside the AC power flow and run to a fixed point, and the downstream quantities (congestion rate, voltage deviation, losses, curtailment) are reported against baselines run in the same loop (Result 2).
- Data and code.** The full pipeline (scenario generators, trainers, figure script) and the generated data and per-seed result files will be released under a permissive license at an anonymous link at submission; no proprietary data is used.

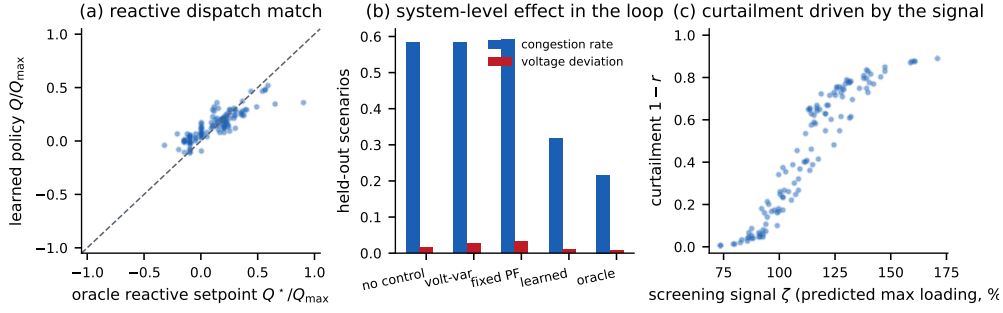


Figure 5: Closed-loop control results on held-out scenarios ($n=120$, seed 0). Left: reactive setpoint applied by the learned policy in the closed loop against the dense-scan oracle setpoint, both normalized by the plant’s reactive headroom; agreement is good where headroom exists. Middle: congestion rate and mean voltage deviation under no control, the IEEE 1547 volt-var droop, fixed 0.95 power factor, the learned policy, and the oracle, all run inside the AC power flow to a fixed point. Right: curtailment $1 - r$ applied by the learned policy against the screening signal ζ ; curtailment concentrates in the scenarios the GNN flags as congested.

policy	held-out		OOD load	
	objective J	congestion (%)	objective J	congestion (%)
no control	0.193	58	0.846	97
IEEE 1547 volt-var	0.317	58	1.652	100
fixed 0.95 PF	0.355	59	1.157	97
learned policy	0.079	31	0.605	98
oracle (dense scan)	0.031	22	0.199	90

Table 2: Mean objective (3) and congestion rate under each policy on the held-out split of the control scenarios and on the OOD load regimes. The learned policy is evaluated in the closed loop to a fixed point, the same loop in which the baselines and the dense-scan oracle are evaluated. The oracle is the per-scenario optimum of the dense scan, so it is the achievable upper bound.

seed	max loading RMSE (%)			congestion acc		
	in-dist	OOD load	N-1 topo	in-dist	OOD load	N-1 topo
0	6.42	10.14	28.29	0.913	0.992	0.984
1	6.20	10.62	32.59	0.908	0.990	0.984
2	6.45	10.74	31.49	0.923	0.994	0.986
3	5.83	10.92	32.83	0.912	0.994	0.984

Table 3: Per-seed GNN screening metrics, showing the variance across seeds that the body reports as means.

seed	held-out J	OOD J	held-out cong (%)	OOD cong (%)
0	0.073	0.594	32	98
1	0.094	0.617	33	98
2	0.073	0.603	31	98
3	0.065	0.596	24	98
4	0.091	0.617	33	98

Table 4: Per-seed closed-loop control metrics for the learned policy.